

Does Hierarchical Summarization Beat Flat Truncation on Long-Form Fiction?

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Abstract

An abstractive summarization models such as BART (Lewis et al., 2020) cap their input at 1024 tokens, forcing a choice when summarizing chapters that exceed this limit: truncate the input, or split it into chunks and summarize hierarchically. We compare flat truncation against a chunk-then-combine hierarchical approach on 40 chapters from the BookSum corpus, using facebook/bart-large-cnn without fine-tuning. Across ROUGE-1/2/L and a proper-noun entity-recall check, we find no statistically significant difference between the two approaches using paired Wilcoxon with all $p > 0.44$, with effect sizes shrinking toward zero as the sample grew from 8 to 40 chapters, indicating a genuine null rather than an underpowered test. We compare two mechanisms: Truncation already keeps the early-chapter content that reference summaries focus on, and hierarchical summarization loses detail by compressing twice.

1 Introduction

Long-form serialized fiction (web novels, light novels) poses a practical summarization problem: chapters are long, plot-dense, and readers who fall behind want to catch up without any long, boring re-reading sessions. Or any "readers" with a short attention span may preferred summary content instead. Sometimes, model have a certain token limit, hence we explore two different mechanisms to cope with any tokens limit and compare them, hierarchical summarization and flat truncation. We came up with the key question: "Does hierarchical (chunk-then-combine) summarization beat flat truncation on Long-Form Fiction".

2 Method

We use facebook/bart-large-cnn, an abstractive encoder-decoder summarizer, loaded directly via AutoModelForSeq2SeqLM. (Wolf et al., 2020) BART's encoder accepts at most 1024 tokens; because the transformer self-attention mechanism

scales quadratically with sequence length (Vaswani et al., 2017), this limit is architectural rather than incidental, and will be the constraint our comparison is built directly on.

Flat truncation (baseline). The chapter is truncated to the first 1024 tokens and summarized in a single pass; all later content is discarded.

Hierarchical. The chapter is split into ~ 1000 -token chunks on sentence boundaries, each chunk is summarized independently, the chunk summaries are concatenated, and that concatenation is summarized once more into the final summary.

3 Data

We evaluate on the cleaned BookSum chapter split (Kryściński et al., 2022). BookSum source texts are public-domain novels from Project Gutenberg; reference summaries were collected from web study-guide sources. This is still a proxy for modern web novels. The domain differs, but the structural (narrative input exceeding the model's context window) as well as general long-form fiction format like light novel or web novels is the same. We sample 40 chapters of at least 1500 words so that chapters exceed the 1024-token limit.

4 Evaluation

We report ROUGE-1/2/L F1 (Lin, 2004) against the human reference summary. ROUGE measures lexical n -gram overlap, not factual correctness; we treat it as a proxy and interpret scores relative to the baseline rather than against some certain absolute thresholds. This is because both conditions run on the same chapters, scores are paired; with n small and no way to verify normality of the differences, we use the non-parametric paired Wilcoxon signed-rank test rather than a t -test. As a complement to ROUGE's lexical bias, we also measure entity recall, which is the fraction of proper nouns in the reference that survive into each generated summary.

5 Results

Table 1 reports mean scores at $n = 40$. The two approaches are within ~ 0.002 on all three ROUGE

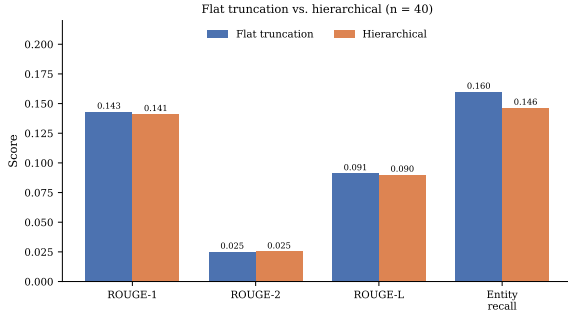


Figure 1: Flat truncation vs. hierarchical across all four metrics ($n = 40$). Scores are near-identical on the three ROUGE metrics; hierarchical is slightly lower on entity recall.

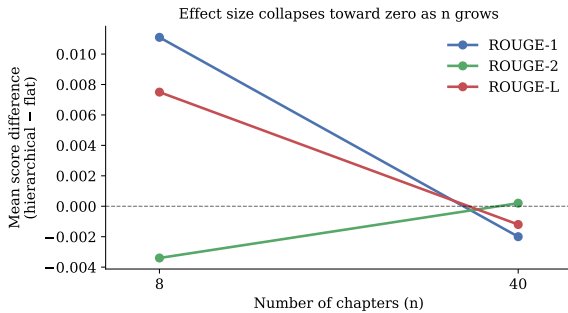


Figure 2: Mean score difference (hierarchical - flat) at $n = 8$ vs. $n = 40$. Differences collapse toward zero as the sample grows, indicating a genuine null rather than an underpowered test.

metrics, and the paired Wilcoxon test finds no significant difference on any metric (ROUGE-1 $p = 0.65$, ROUGE-2 $p = 0.46$, ROUGE-L $p = 0.44$). Figure 1 shows the near-identical scores visually.

Crucially, the effect sizes *shrank* as the sample grew from 8 to 40 chapters

(Figure 2): differences that were ± 0.01 at $n = 8$ collapsed toward zero at $n = 40$. A power-limited null would instead show a stable gap that merely failed to reach significance; the shrinkage indicates a *genuine* null.

The one consistent signal is entity recall, where hierarchical is lower in both runs (0.146 vs. 0.160 at $n = 40$), consistent with entities that appear in a single chunk being dropped during the second summarization pass.

6 Discussion and Limitations

Two reasons explain the null. First, hierarchical summarization only helps if the truncation throws away content the reference summary actually cares about. We hypothesise that BookSum reference summaries focus on early-chapter setup, exactly the part truncation keeps so truncation loses little that ROUGE rewards, leaving hierarchical little to gain. Confirming this would require mea-

Metric	Flat truncation	Hierarchical
ROUGE-1	0.1431	0.1411
ROUGE-2	0.0251	0.0253
ROUGE-L	0.0912	0.0900
Entity recall	0.1599	0.1464

Table 1: Mean scores over 40 BookSum chapters. No ROUGE difference is significant (paired Wilcoxon, all $p > 0.44$).

suring where reference-relevant content often falls within each chapter. Second, although hierarchical summarization sees more of the chapter, it summarizes twice (even more on higher token data), and each pass loses detail. This shows up in its lower entity recall, where it drops more character and place names. The extra coverage and the lost detail roughly cancel, which is why neither method wins on ROUGE.

Limitations. BookSum is a domain proxy, not actual web novels, so results may not transfer well to modern web-novel style where more "Internet" term were used. With $n = 40$ the study is modestly powered and detects sizeable effects. ROUGE is purely lexical and cannot credit correct paraphrases, but a semantic metric (e.g. BERTScore) (Zhang et al., 2020) would strengthen the evaluation. We also mention that Entity recall uses a capitalization-based proper-noun finder function rather than a trained NER model, so it over- and under-counts entities, therefore it should only be considered as an estimation.

7 Conclusion

To conclude this project, on BookSum chapters, hierarchical summarization does not outperform flat truncation, and is marginally worse at retaining named entities. The benefit of chunking appears to depend on where reference-relevant information sits in the chapter.

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8 Contributions and AI Assistance

We used Large Language Models for debugging, code fixing and explanation of complex concepts that was NOT covered in the curriculum.

We were responsible for every section in this report and the following github repo.

<https://github.com/NamTrank5/chap-summary>